

Merlin Sparrow Plus

Multifunctional Pocket Hardware Debugging Assistant

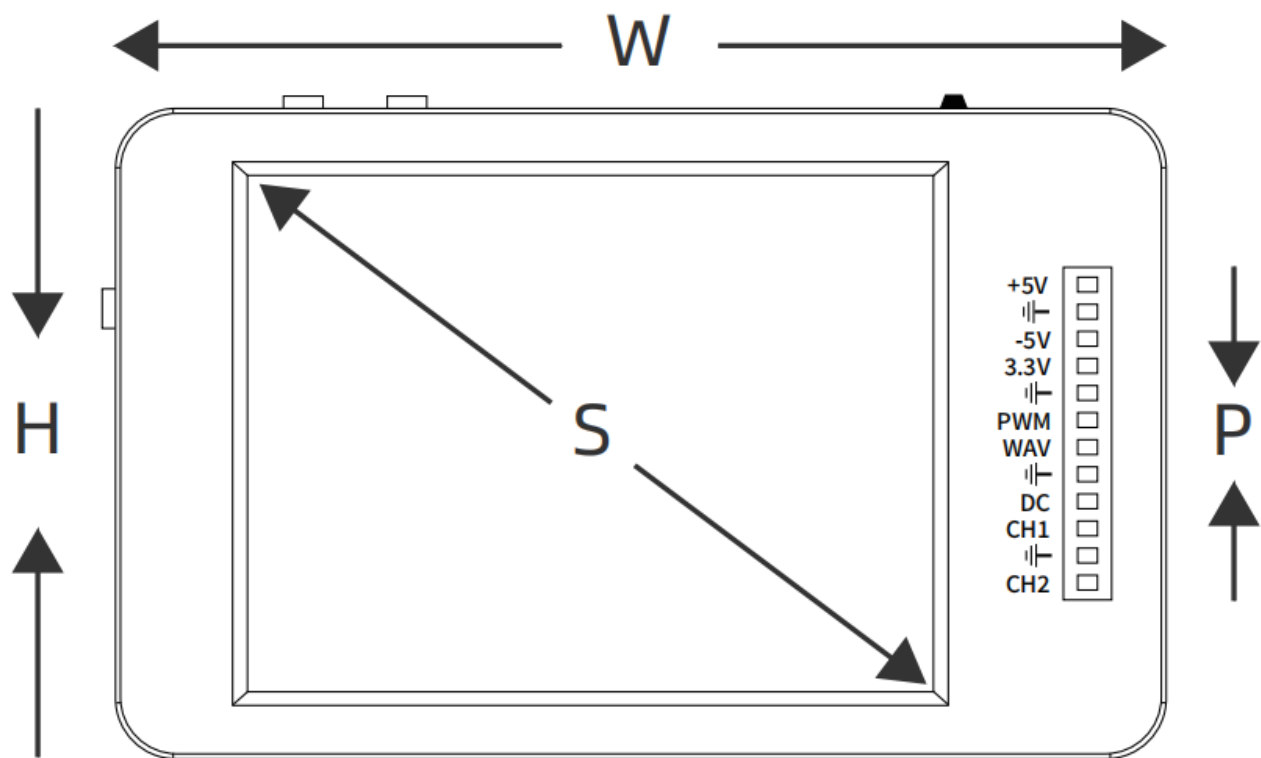
Product Information

- **Full name:** Multifunctional Pocket Hardware Debugging Assistant
- **Nickname:** Merlin Sparrow Plus
- **Development team:** Suzhou Sidepu Information Technology Co., Ltd.
- **Core chip:** STM32G473

Package Contents

- Merlin Sparrow Plus x1
- Dual-female 12-wire Dupont cable x1
- USB Type-C cable x1
- User manual x1
- Premium packaging box x1
- 2.54 mm pitch 40-pin header x1

Dimensions and Physical Specifications



Parameter	Value
H	62 mm

Parameter	Value
W	100 mm
S	83 mm
P	32 mm
Thickness	15 mm
Device net weight	90 g
Series color	White

Specifications

Mode	Parameter	Specification	Port
Signal Generator	Output waveform	Sin/Tri/Sqr	WAV
Signal Generator	Output frequency range	[0.1 Hz, 100 kHz]	WAV
Signal Generator	Output amplitude range	[-4 V, 4 V]	WAV
Signal Generator	DC offset range	[-4 V, 4 V]	WAV
PWM Signal	Output voltage range	[0 V, 3.3 V]	PWM
PWM Signal	Output frequency range	[0.1 Hz, 5 MHz]	PWM
PWM Signal	Output duty cycle	[5%, 95%]	PWM
DC Voltage Source	Output voltage range	[-4 V, 4 V]	DC
DC Voltage Source	Voltage adjustment accuracy	±0.5%	DC
Oscilloscope	Maximum input voltage	20 Vpp	CH1 / CH2
Oscilloscope	Sampling rate	5 Msps	CH1 / CH2
Oscilloscope	Analog bandwidth	500 kHz	CH1 / CH2
Oscilloscope	Input impedance	1 MΩ	CH1 / CH2
Oscilloscope	Coupling	DC coupling	CH1 / CH2
Power Supply	+5 V	100 mA	+5 V
Power Supply	-5 V	100 mA	-5 V
Power Supply	+3.3 V	100 mA	+3.3 V
Other Parameters	Display size	3.2 in	—
Other Parameters	Screen resolution	320 × 240	—
Other Parameters	Battery capacity	1200 mAh	—
Other Parameters	Battery standby time	8 hours	—

Additional Tips

About Signal Generation

The linearity of the output triangle-wave signal will gradually decrease at 500 kHz or above.

When the WAV port is used to generate a square-wave signal, the rising and falling edges become noticeably slower when the output frequency exceeds 100 kHz. Therefore, if a high-frequency square wave is required, it is recommended to use the PWM port.

The DC port can provide a relatively high-accuracy DC reference output. However, its source and sink currents are relatively small, so it is not recommended for use as a load-driving port.

About Signal Measurement

CH1 and CH2 can both be used as oscilloscope channels at the same time.

CH1 and CH2 use DC coupling by default during measurement.

When used as an oscilloscope, the device can also be connected to a computer and supports functions such as data export and waveform screenshots.

When the measured signal is approximately 30 V peak-to-peak, clipping at the top and bottom will occur, resulting in inaccurate measurement results. Therefore, make sure the measured signal amplitude remains within a reasonable range.

Precautions

1. This product has a free limited warranty period of 365 days from the date of purchase.
2. The DC signal output impedance is not high impedance. Do not short it to another power supply, otherwise the device may be damaged.
3. This product is not a children's toy. Children under 14 years old should use it under the supervision of a guardian.
4. If a power adapter is used to power the device, the adapter must have proper third-party certification. Do not use uncertified adapters or cables.
5. The cables for the device ports are specially customized. If they are lost or damaged, purchase replacements from the original manufacturer. Non-customized cables may cause functional errors.
6. Forcibly opening the device will cause permanent structural damage. If there is a product problem, contact the original manufacturer's technical support promptly.

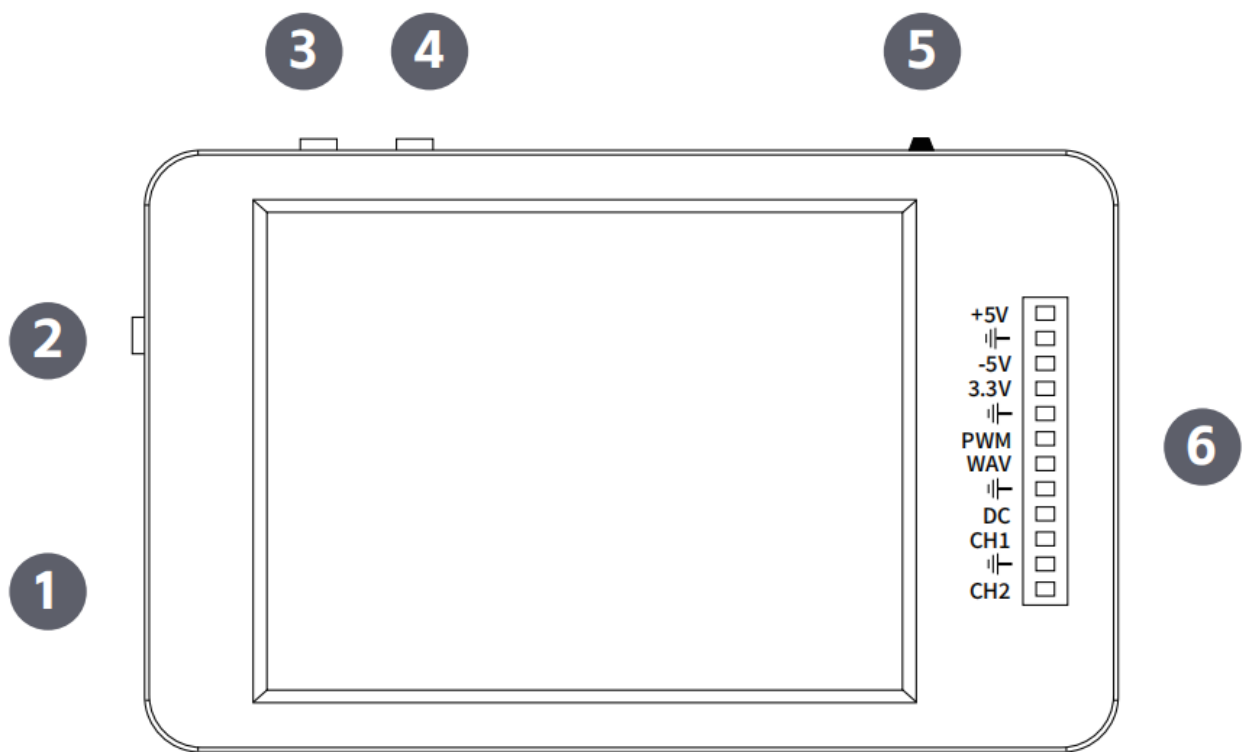
Free Repair Is Not Provided in the Following Cases

The company's products are covered by a limited warranty within the warranty period. Device failures or damage under the following circumstances are not covered by the limited warranty, and the company is not obligated to provide free repair:

1. The device or component is beyond the valid warranty period.

2. Device failure or damage caused by incorrect installation, storage, or use that does not follow the requirements of the manual.
3. Device failure or damage caused by disassembly by an organization or person not authorized by the company.
4. Device failure or damage caused by accidental factors or human causes, including but not limited to operating errors, scratches, transportation, and impacts.
5. Device failure or damage caused by force majeure such as natural disasters, including earthquakes and fires.

Controls and Ports



1. Type-C USB Port
2. Power Button
3. Function Switch Button
4. Menu Selection Button
5. Adjustment Switch
6. Debug Port

Main Functions

1. Type-C USB Port

Operation:

Connect to a 5 V power supply or a computer USB port.

Functions:

- Charge the device
- Update device firmware

2. Power Button

Operation:

Long press / short press.

Functions:

- Short press to power on.
- Long press for 3 seconds and release to power off.

3. Function Switch Button

Operation:

Short press.

Functions:

- Signal generator
- Oscilloscope time-domain signal measurement
- Spectrum measurement
- High-accuracy DC offset signal
- PWM signal output

4. Menu Selection Button

Operation:

Short press.

Function:

Under the current function, press this button to switch between different parameter adjustment selections.

5. Adjustment Switch

Operation:

Move left / move right / press.

Functions:

- **Left:** Decrease or move left
- **Right:** Increase or move right
- **Press:** Confirm and switch to the next digit

6. Debug Port

Operation:

Connect the 12-wire Dupont cable.

Functions:

- **+5 V, -5 V, +3.3 V:** Fixed power output
- **PWM (OUT):** Adjustable duty cycle
- **WAV (OUT):** Triangle wave, sine wave, square wave
- **DC (OUT):** High-accuracy DC offset
- **CH1 (IN):** Oscilloscope channel 1
- **CH2 (IN):** Oscilloscope channel 2

Merlin Sparrow Plus Supports Secondary Firmware Development

To perform secondary firmware development, the device must be placed into DFU development mode. Follow the steps below.

DFU Mode

STEP 1

Make sure the device is powered off.

STEP 2

Connect the device to the computer's USB port while keeping the device powered off.

STEP 3

Press and hold button **5**.

STEP 4

Press and hold button **2**. At this point, the Device Manager should display the corresponding indication.

STEP 5

Keep button **2** pressed until firmware programming is complete.

Note

When a DFU device appears in the computer's Device Manager, button **5** can be released.

The main chip is an STM32G473 microcontroller. The IDE tool (CUBE) can be downloaded from the ST official website for firmware development.